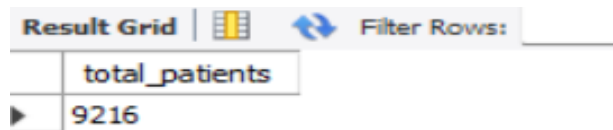


HOSPITAL EMERGENCY ROOM SQL QUERY

PATIENT KPI'S

-- 1) Total number of Patient

```
SELECT COUNT(*) AS total_patients  
FROM hospital_emergency_room;
```

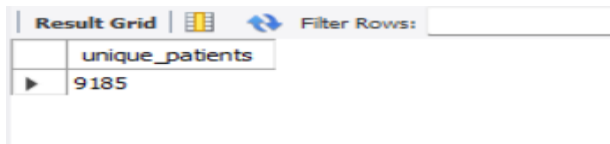


The screenshot shows a SQL query result grid with a single column labeled 'total_patients' and a single row with the value '9216'. The grid has a 'Result Grid' tab and a 'Filter Rows' button.

total_patients
9216

-- 2) Total number of Unique Patient

```
SELECT COUNT(DISTINCT patient_name) AS unique_patients  
FROM hospital_emergency_room;
```

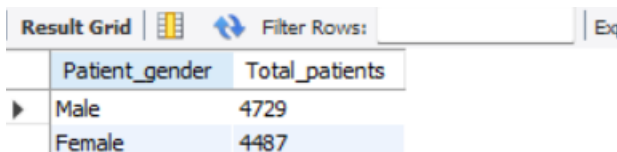


The screenshot shows a SQL query result grid with a single column labeled 'unique_patients' and a single row with the value '9185'. The grid has a 'Result Grid' tab and a 'Filter Rows' button.

unique_patients
9185

-- 3) Total number of Male and Female;

```
SELECT Patient_gender, COUNT(*) AS Total_patients  
FROM hospital_emergency_room  
GROUP BY Patient_gender;
```



The screenshot shows a SQL query result grid with two columns: 'Patient_gender' and 'Total_patients'. There are two rows: 'Male' with a count of 4729, and 'Female' with a count of 4487. The grid has a 'Result Grid' tab, a 'Filter Rows' button, and an 'Export' button.

Patient_gender	Total_patients
Male	4729
Female	4487

-- 4) Age group Distribution;

```

SELECT Patient_age_group, COUNT(*) AS Total_patients
FROM hospital_emergency_room
GROUP BY Patient_age_group
ORDER BY Total_patients DESC;

```

Result Grid   Filter Rows:		
	Patient_age_group	Total_patients
▶	20-29	1207
	30-39	1191
	0-9	1176
	10-19	1160
	60-69	1150
	50-59	1147
	40-49	1137
	70-79	1048

Admission Analysis

-- 1) Total no. of admitted and not admitted patient;

```

SELECT Patient_admission_status,
COUNT(*) AS total
FROM hospital_emergency_room
GROUP BY Patient_admission_status;

```

Result Grid   Filter Rows:		
	Patient_admission_status	total
▶	Not Admitted	4604
	Admitted	4612

-- 2) Admission rate;

```

SELECT
ROUND(
SUM(CASE WHEN Patient_admission_status='Admitted' THEN 1 ELSE 0
END)*100/COUNT(*),2

```

) AS admission_rate

FROM hospital_emergency_room;

Result Grid		Filter Rows:	E
	admission_rate		
▶	50.04		

Patient Visit Trend

-- 1) Daily patient visit

SELECT Patient_admission_date,

COUNT(*) AS patients

FROM hospital_emergency_room

GROUP BY Patient_admission_date

ORDER BY Patient_admission_date;

Result Grid		Filter Rows:	Export
	Patient_admission_date	patients	
▶	2023-04-01	19	
	2023-04-02	13	
	2023-04-03	14	
	2023-04-04	9	
	2023-04-05	19	
	2023-04-06	14	
	2023-04-07	11	
	2023-04-08	22	
	2023-04-09	12	
	2023-04-10	13	
	2023-04-11	17	
	2023-04-12	30	
	2023-04-13	13	
	2023-04-14	21	
	2023-04-15	12	

(Consist 2 year data)

-- 2)Monthly Patient visit

```

SELECT
MONTHNAME(Patient_admission_date) AS month,
COUNT(*) AS patients
FROM hospital_emergency_room
GROUP BY MONTH(Patient_admission_date),MONTHNAME(Patient_admission_date)
ORDER BY MONTH(Patient_admission_date);

```

Result Grid |   Filter Rows: _____

month	patients
January	513
February	431
March	506
April	948
May	999
June	991
July	952
August	1024
September	935
October	964
November	464
December	489

-- 3) Day wise visit

```

SELECT
DAYNAME(Patient_admission_date) AS day_name,
COUNT(*) AS patients
FROM hospital_emergency_room
GROUP BY DAYNAME(Patient_admission_date);

```

Result Grid   Filter Rows:

day_name	patients
Wednesday	1260
Saturday	1377
Thursday	1332
Sunday	1318
Monday	1314
Friday	1310
Tuesday	1305

TIME ANALYSIS

-- 1)Patients by Hour:

SELECT

HOUR(Patient_admission_time) AS hour,

COUNT(*) AS patients

FROM hospital_emergency_room

GROUP BY HOUR(Patient_admission_time)

ORDER BY hour;

Result Grid		Filter Rows:	Result Grid		Filter Rows:
hour	patients		hour	patients	
0	406		9	388	
1	372		10	349	
2	376		11	403	
3	385		12	366	
4	384		13	410	
5	393		14	368	
6	375		15	394	
7	415		16	378	
8	386		17	359	
9	388		18	370	
10	349		19	383	
11	403		20	372	
12	366		21	376	
13	410		22	372	
14	368		23	436	

-- 1)Busiest Hour:

SELECT

```

    HOUR(Patient_admission_time) AS hour,
    COUNT(*) AS patients
FROM hospital_emergency_room
GROUP BY hour
ORDER BY patients DESC
LIMIT 1;

```

Result Grid			Filter Rows:
	hour	patients	
▶	23	436	

Wait Time Analysis

-- 1) Average waiting time

```

SELECT
ROUND(AVG(Patient_waittime),2) AS average_wait
FROM hospital_emergency_room;

```

Result Grid		Filter Rows:
	average_wait	
▶	35.26	

-- 2) Maximum waiting time

```

SELECT
MAX(Patient_waittime) AS max_wait
FROM hospital_emergency_room;

```

Result Grid		Filter Rows:
	max_wait	
▶	60	

-- 3) Patient seen within 30 minutes

```

SELECT
COUNT(*) AS patients_seen_within_30
FROM hospital_emergency_room
WHERE Patient_waittime<=30;

```

	patients_seen_within_30
▶	3749

-- 4) Percentage of Patient seen within 30 minutes

```

SELECT
ROUND(
SUM(CASE WHEN Patient_waittime<=30 THEN 1 ELSE 0 END)*100/COUNT(*),2
) AS percentage
FROM hospital_emergency_room;

```

Result Grid  Filter Rows:	
	percentage
▶	40.68

Department Referral Analysis

-- 1) Patients by Department:

```

SELECT
department_referral,
COUNT(*) AS total_patients
FROM hospital_emergency_room
GROUP BY department_referral

```

ORDER BY total_patients DESC;

department_referral	total_patients
None	5400
General Practice	1840
Orthopedics	995
Physiotherapy	276
Cardiology	248
Neurology	193
Gastroenterology	178
Renal	86

-- 2) Top 5 Departments:

SELECT

department_referral,

COUNT(*) AS total_patients

FROM hospital_emergency_room

GROUP BY department_referral

ORDER BY total_patients DESC

LIMIT 5;

	department_referral	total_patients
►	None	5400
	General Practice	1840
	Orthopedics	995
	Physiotherapy	276
	Cardiology	248

Satisfaction Analysis

-- 1) Average Satisfaction Score;

```
SELECT
ROUND(AVG(patient_satisfaction_score),2)
FROM hospital_emergency_room;
```

	ROUND(AVG(patient_satisfaction_score),2)
▶	1.36

-- 2) Satisfaction by Department;

```
SELECT
department_referral,
ROUND(AVG(patient_satisfaction_score),2) AS avg_score
FROM hospital_emergency_room
GROUP BY department_referral
ORDER BY avg_score DESC;
```

	department_referral	avg_score
▶	Gastroenterology	1.76
	Physiotherapy	1.50
	Cardiology	1.47
	Neurology	1.45
	Orthopedics	1.42
	General Practice	1.38
	None	1.32
	Renal	1.22

DEMOGRAPHIC ANALYSIS

-- 1) Patients by Race:

```
SELECT patient_race,
COUNT(*) AS total
FROM hospital_emergency_room
```

GROUP BY patient_race;

Result Grid			Filter Rows:	Exp
	patient_race	total		
▶	White	2571		
	Native American/Alaska Native	498		
	African American	1951		
	Asian	1060		
	Two or More Races	1557		
	Pacific Islander	549		
	Declined to Identify	1030		

-- 2) Patients by Gender and Age Group:

```
SELECT patient_gender, patient_age_group,  
COUNT(*) AS total  
FROM hospital_emergency_room  
GROUP BY patient_gender, patient_age_group;
```

Result Grid				Filter Rows:	Exp
	patient_gender	patient_age_group	total		
▶	Male	60-69	588		
	Male	0-9	586		
	Female	50-59	547		
	Female	20-29	581		
	Male	50-59	600		
	Female	60-69	562		
	Female	40-49	574		
	Female	70-79	524		
	Female	10-19	543		
	Male	40-49	563		
	Male	30-39	625		
	Female	0-9	590		
	Male	20-29	626		
	Male	10-19	617		
	Female	30-39	566		

PEAK DAYS

-- 1) Top 10 busiest days:

```
SELECT patient_admission_date,
```

```

COUNT(*) AS patient
FROM hospital_emergency_room
GROUP BY patient_admission_date
ORDER BY patient DESC
LIMIT 10
;

```

patient_admission_date	patient
2023-04-12	30
2024-08-01	28
2024-05-21	28
2023-12-15	27
2023-09-15	26
2023-06-22	26
2023-06-01	26
2024-03-07	26
2023-10-18	25
2024-08-17	25

-- Rank Departments by Patient Count:

```

SELECT department_referral,
COUNT(*) AS patient,
RANK() OVER(ORDER BY COUNT(*) DESC) AS ranking
FROM hospital_emergency_room
GROUP BY department_referral;

```

	department_referral	patient	ranking
▶	None	5400	1
	General Practice	1840	2
	Orthopedics	995	3
	Physiotherapy	276	4
	Cardiology	248	5
	Neurology	193	6
	Gastroenterology	178	7
	Renal	86	8

-- Average count of daily patient;

```

SELECT
    ROUND(AVG(daily_patients), 2) AS average_daily_patient_count
FROM (
    SELECT
        patient_admission_date,
        COUNT(*) AS daily_patients
    FROM hospital_emergency_room
    GROUP BY patient_admission_date
) AS daily_count;

```

	average_daily_patient_count
▶	15.92

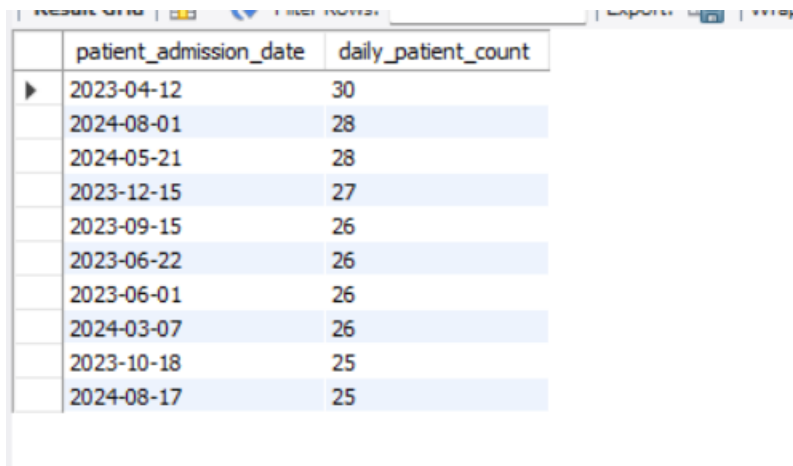
-- Patient for each day:

```

SELECT
    patient_admission_date,

```

```
COUNT(*) AS daily_patient_count
FROM hospital_emergency_room
GROUP BY patient_admission_date
ORDER BY daily_patient_count DESC
LIMIT 10;
```



	patient_admission_date	daily_patient_count
▶	2023-04-12	30
	2024-08-01	28
	2024-05-21	28
	2023-12-15	27
	2023-09-15	26
	2023-06-22	26
	2023-06-01	26
	2024-03-07	26
	2023-10-18	25
	2024-08-17	25